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World's First Rigless Fishbone Needle Retrieval Boosts Oil and Gas Output

I. Soliman, A. Bin Sumaida, and E. Al Shamisi, ADNOC Onshore; H. Al Shehhi, Abu Dhabi National Oil Company; M. A. Hammadi and N. Alhosani, ADNOC Onshore; K. Al Hadidy, Abu Dhabi National Oil Company; F. Abdussalam and A. Reshetnikov, ADNOC Onshore; Y. Faizov, Abu Dhabi National Oil Company; N. Almteiri, A. F. Ali, and M. Shafei, ADNOC Onshore; M. Malak, S. Rashid, and W. Chehabi, Fishbones AS; E. Franco, Schlumberger; A. Elattar and S. Seoudy, SLB

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Abstract

The objective of this paper is to document and compare the performance impact of retrieving fishbone needles using a traditional drilling rig versus a newly developed rigless method, utilizing a coiled tubing unit equipped with a tower. The study highlights operational efficiencies, production outcomes, and the broader implications of adopting a rigless approach for similar interventions.

The approach began with 3D modeling to simulate the rig-up layout and tubing forces simulations to select the optimal Coiled Tubing (CT) pipe and define cutting parameters. Equipment integration was validated through multiple meetings and yard trials between the fishbone and CT systems. Deployment of the 70-ft bottomhole assembly was achieved using a specialized CT tower and pressure control equipment, eliminating the need for mechanical lifting under live well conditions. During operations, up to three fishbone subs were cleared per run, reducing overall intervention time from 33 to 10 days compared to traditional rig-based methods.

Following a detailed technical collaboration between the operator and a coiled tubing service provider, a novel solution was successfully deployed under live-well conditions. This innovative approach was implemented for the first time globally across two wells. In the first well, 40 fishbone subs were retrieved, resulting in gas production rates 300% higher than expected. In the second well, an oil producer, 30 subs were retrieved, achieving oil rates 400% above initial forecasts. The method significantly reduced operational time compared to traditional rig-based interventions and demonstrated major production and economic benefits. These results mark a significant advancement in live well intervention capabilities, providing a scalable and efficient solution for future fishbone retrieval operations and setting a new benchmark in well completion and production enhancement strategies.

This paper presents the first global application of a novel deployment method for retrieving needles from fishbone subs during well completion. A performance analysis comparing the benefits of a rigless intervention is also presented, demonstrating the return on investment and reinforcing the method's economic and operational value.

UAE Oilfield Industry Overview

The United Arab Emirates (UAE) continues to advance its strategic goal of boosting crude oil production capacity to 5 million barrels per day (b/d) by 2027, accelerating its original 2030 target. This ambition, driven by anticipated global demand and national energy strategies, is accompanied by a strong emphasis on optimizing operational expenditure and maximizing resource recovery (EIA, 2024).

To meet these objectives, the widespread deployment of horizontal wells in tight carbonate reservoirs has become standard. These complex reservoirs, often exhibiting permeability below 10 mD and poor vertical communication ($Kv/Kh < 0.5$), require innovative completion and stimulation strategies to ensure effective hydrocarbon production (Rachapudi et al., 2020; Subba Ramarao et al., 2024). However, traditional techniques such as matrix acidizing or multistage fracturing are frequently constrained by placement inefficiencies and formation heterogeneity.

To overcome these limitations, operators have progressively adopted advanced well architectures, including fishbone completions— multi-lateral extensions designed to maximize reservoir contact and improve sweep efficiency. While these configurations improve production, they also pose unique intervention challenges, particularly during the cleanup or retrieval of completion components post-stimulation.

Historically, such interventions required rig-based operations, incurring substantial cost, time, and logistical complexity. However, the emergence of rigless solutions—most notably, CT deployed via specialized towers—has created a new paradigm for live well interventions. These methods significantly reduce non-productive time and expand the feasibility of performing complex retrievals under pressure-controlled environments.

This evolution in completion and intervention strategy reflects the UAE's broader shift toward scalable, high-efficiency technologies that support both production growth and sustainable field development

Fishbone Completion Technology Overview

As part of its continued efforts to enhance recovery from tight carbonate reservoirs, the operator has adopted fishbone completion technology—a novel uncemented liner system engineered to increase reservoir contact and improve vertical connectivity in formations with low permeability and poor interlayer communication (Subba Ramarao et al., 2024). This technique is particularly suited to the UAE's challenging carbonate environments.

The system integrates jetting-enabled subs equipped with high-strength metallic needles. During stimulation, these needles are hydraulically deployed through acid jetting and penetrate the formation up to 40 feet, effectively bypassing near-wellbore damage and establishing conductive pathways across multiple reservoir layers and natural fractures. This geometry enhances vertical and radial flow, optimizing the drainage area and enabling more uniform stimulation along the horizontal lateral.

Fishbone Stimulation Process Overview

The fishbone completion process comprises three main stages:

Stage 1: Installation of the Fishbone Assembly

- A conventional drilling rig is used to run the uncemented fishbone liner and upper completion into the wellbore.
- Fishbone subs, each outfitted with four acid-activated jetting needles, are spaced along the liner per stimulation design.
- The assembly includes open-hole anchors and an acid-activated shoe designed to create a pressure-sealed environment at target depth.

- Once landed, the liner hanger and top packer are set to anchor the completion in place.

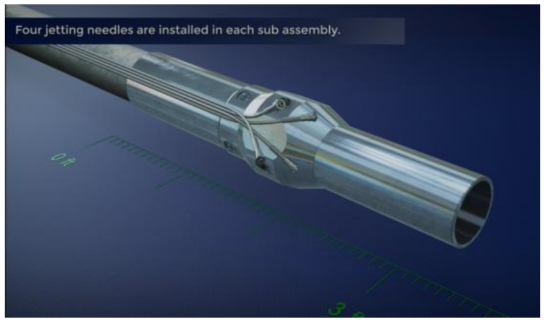

Figure 1—The fishbone sub assembly[13].

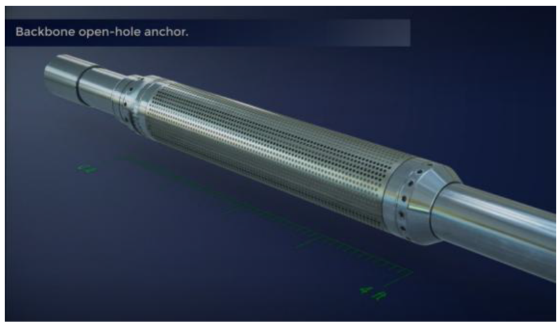

Figure 2—The backbone openhole anchor[13].

Figure 3—The fishbone liner hanger[13].

Stage 2: Jetting Needle Activation and Acid Stimulation

- Low-inhibited acid is pumped through the system. Contact with the acid initiates a chemical reaction in the activation shoe, forming a sealed chamber.
- Pressure is applied to set the open-hole anchors, immobilizing the liner.
- At a predefined pressure threshold, the needles are released and simultaneously jet acid as they extend into the reservoir matrix.
- This process ensures targeted acid placement and effective stimulation of multiple reservoir layers, enhancing reservoir connectivity.

Figure 4—The fishbone shoe[13].

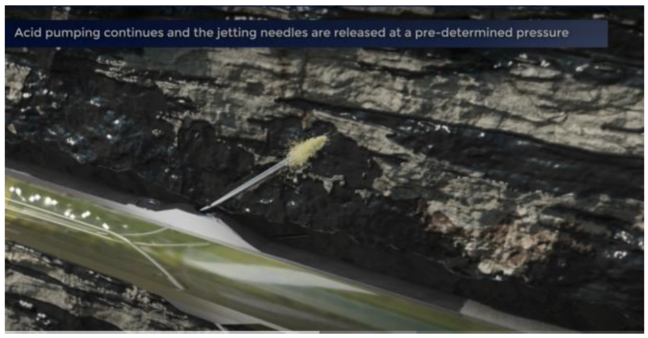

Figure 5—The above figure illustrates the acid jetting through the fishbone needles[13].

Stage 3: Needle Retrieval and Upper Completion

- Traditionally, needle retrieval is performed using drill pipe and a mechanical fishing tool to cut and extract the metallic tails left inside the liner ID.
- This step requires pumping kill fluid and deploying loss control material to prevent losses, posing operational and reservoir- related risks.
- Only after retrieval is completed can the upper completion be installed and the well brought online.
- Hydrocarbon flow is then directed through the needle annuli into the mother bore and onward through either production valves or inflow control devices (ICDs), depending on the configuration.

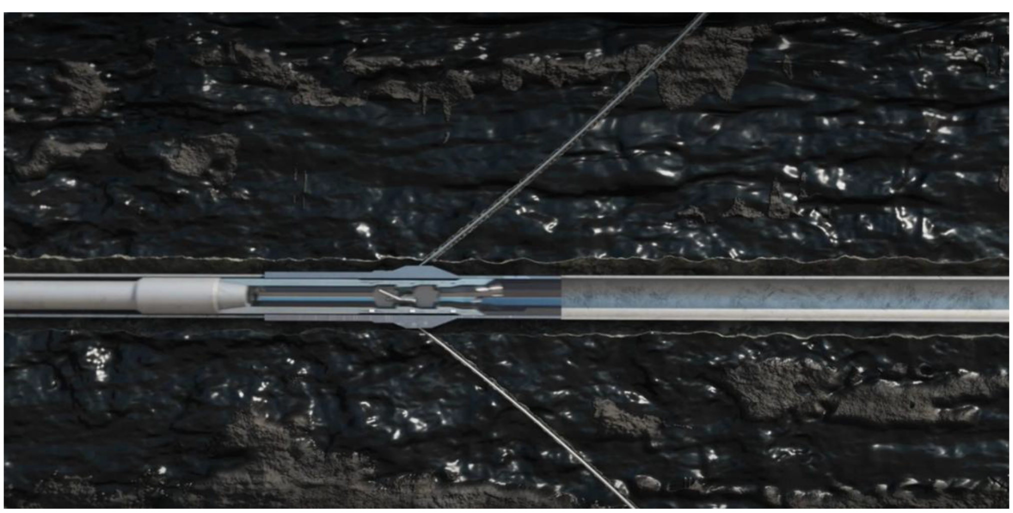

Figure 6—The above figure illustrates the needles leftover retrieval process using a drill pipe and the cutting head[13].

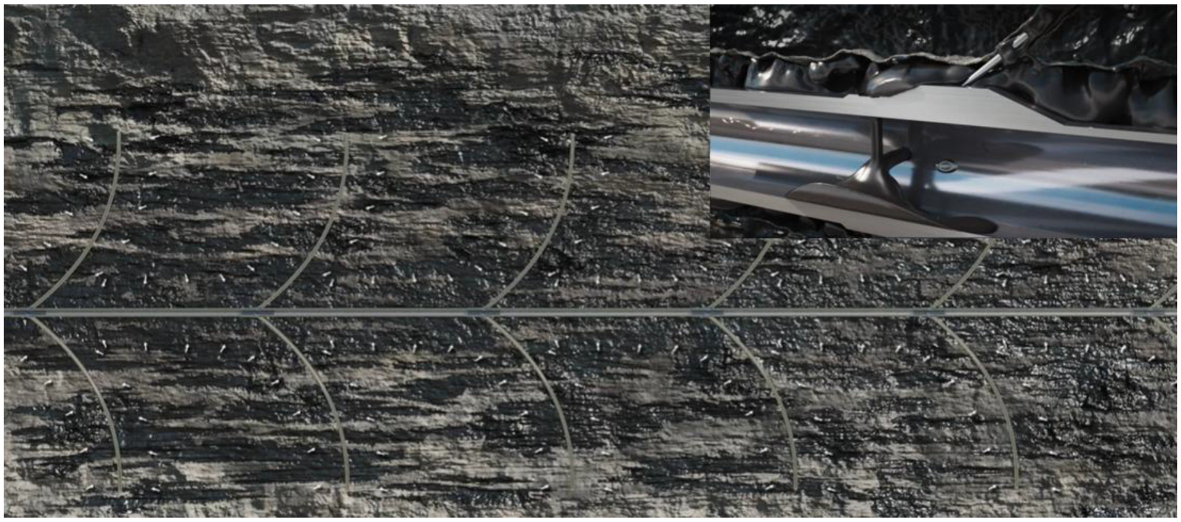

Figure 7—The above figure illustrates oil flow path from the reservoir into the completion[13].

While the stimulation stage of the fishbone system is highly efficient, **the needle retrieval process has historically been a bottleneck**, requiring rig-based intervention under dead-well conditions. This dependency not only adds cost and time but also introduces operational constraints—especially in remote or rig-constrained environments. The case study presented in this paper explores an industry-first solution to this challenge, demonstrating how **a coiled tubing-based, rigless approach** can be safely and effectively used to retrieve fishbone needles under live-well conditions, without mechanical lifting.

The Challenge

While high oil prices improve profitability across the industry, a significant portion of upstream spending continues to be directed toward technical engineering services. According to Song et al. (2021), this trend underscores the need for greater operational efficiency and adoption of innovative technologies—particularly in mature fields where production decline and reservoir complexity intensify cost pressures. In the UAE, tight carbonate formations are among the most challenging, requiring carefully engineered stimulation and completion techniques to achieve target productivity (Rachapudi et al., 2020).

However, as demonstrated in other energy revolutions—such as the U.S. shale boom—innovation alone does not guarantee efficiency. It was the integration of multiple technologies (e.g., hydraulic fracturing with directional drilling) that unlocked transformative value (Zhang et al., 2022; Gong, 2017). Importantly, the early phases of adopting new technologies often come with operational inefficiencies, learning curves, and unforeseen risks. Over time, the combination of maturing processes and adaptive deployment strategies drives tangible improvements.

This principle held true during the operator's initial deployment of fishbone completions. The early phase used a conventional open- system approach: a drilling rig and jointed pipe were employed to retrieve the post-stimulation metallic needles from the liner before running the upper completion. While functionally successful, these operations faced major well control challenges. Unexpected formation behavior led to insufficient surface returns, triggering severe circulation losses. More than 30,000 barrels of loss-control material were pumped into the formation—compromising reservoir integrity and complicating future production.

In at least one case, the well had to be killed using heavy mud to restore control, delaying cleanout operations and potentially damaging the newly stimulated zones. Rachapudi et al. (2024) identified well control, circulation loss, and reservoir stability as three critical challenges during early field trials.

These outcomes made it clear that **a new approach was required,** one that would eliminate the need for rig-based operations, preserve reservoir integrity, and safely perform critical intervention steps such as needle retrieval **under live-well conditions**.

The Innovation Process

The operational challenges encountered during early fishbone deployments—especially circulation losses, well control complications, and the use of kill mud—underscored the urgent need for process innovation. In response, the engineering team applied the **subtraction technique** from the *Systematic Inventive Thinking* (SIT) methodology (Boyd and Goldenberg, 2013), which encourages "inside-the- box" thinking by deliberately removing a core system component to uncover new possibilities.

In this case, the drilling rig—the costliest and most complex element of the workflow—was identified as the component to subtract. This led to the conceptualization of a **rigless intervention strategy** for retrieving fishbone needles under live-well conditions. However, deploying a long, mechanical bottomhole assembly (BHA) without a rig introduced a set of technical challenges requiring innovative engineering, detailed risk management, and cross-functional collaboration.

Evaluating Deployment Alternatives

Two rigless deployment strategies were considered:

1. **Pressure Deployment via Slickline or Wireline**
   This method proposed lowering the BHA into the well, using slickline or wireline, supported by a crane. While technically feasible, it required over 23 separate lifting operations above a live well, significantly increasing the operational risk profile. Due to these unacceptable safety concerns, the method was discarded.
2. **Coiled Tubing with Specialized Tower Support**
   A second, more robust approach involved the use of a **CT unit equipped with a hydraulically supported tower**. This solution eliminated the need for mechanical lifting over the wellhead during operations, vastly improving safety and repeatability.

Key Components of the Rigless Solution

a. **Coiled Tubing Tower Integration**
   The CT tower replaced most crane operations during intervention, except for rig-up and rig-down. A hydraulic support frame allowed the deployment and retrieval of ~70-ft fishing BHAs without any mechanical lifting over the live well. The fishing assembly incorporated an internal cutting basket and flapper system designed to cut and capture needle tails inside the liner.

b. **Customized Pressure Control Equipment (PCE)**
   To maintain pressure integrity and expedite operations, the team designed a PCE system featuring **quick-test subs (QTS)** for faster pressure testing and streamlined BHA cycling. This also reduced exposure time and crane use during red-zone operations.

c. **Updated Operational Procedures & Contingency Planning**
   New operating procedures were developed to support this rigless workflow, including detailed contingency plans to address stuck BHAs, emergency disconnects, and tool retrieval. The CT tower setup allowed for pressure testing in a sealed chamber before each run (red zone), followed by BHA deployment via CT (blue zone), cutting operations, and retrieval for unloading the needle tails at surface.

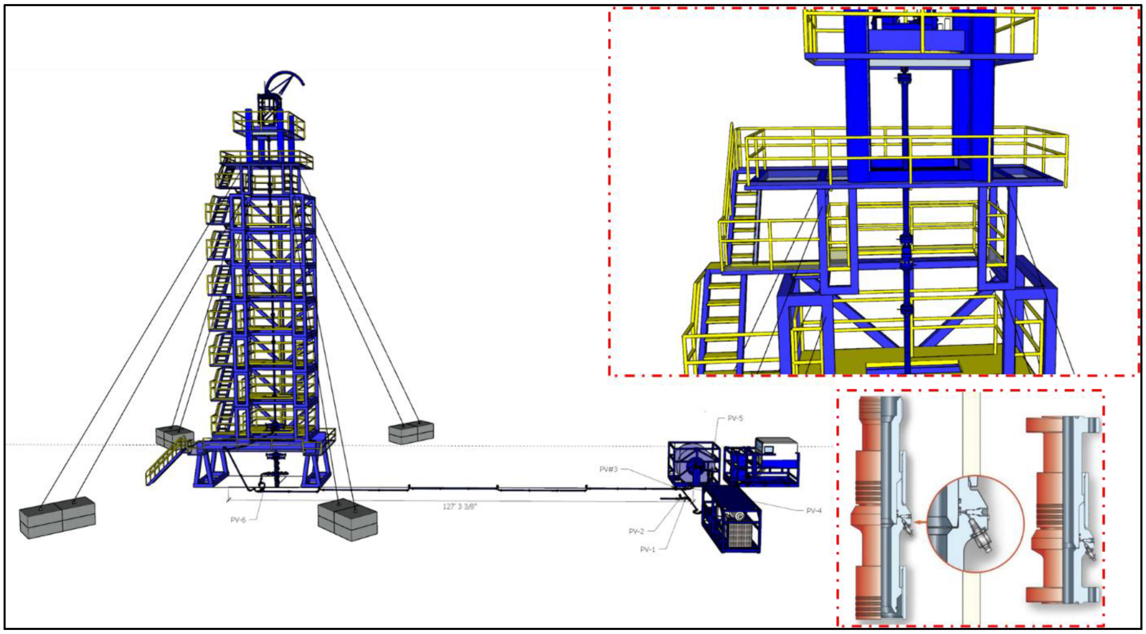

Figure 8—The CT Tower rigless approach was deemed the optimum solution. The red boxes zoom to the PCE indicating the position of the QTS at the upper side of the risers[1].

Results of Innovation

This approach was field-tested and demonstrated robust performance. The team achieved up to **two full cycles per day**, with each cycle retrieving up to **four fishbone subs (16 needle tails)**. Compared to the conventional rig-based approach, this represented a substantial gain in operational efficiency and a significant reduction in HSE exposure, NPT, and formation damage risk.

By removing the rig from the workflow and engineering a CT-based live-well solution, the team introduced a scalable innovation that redefined the intervention strategy for fishbone completions—delivering value through **cost savings, technical reliability, and enhanced safety**.

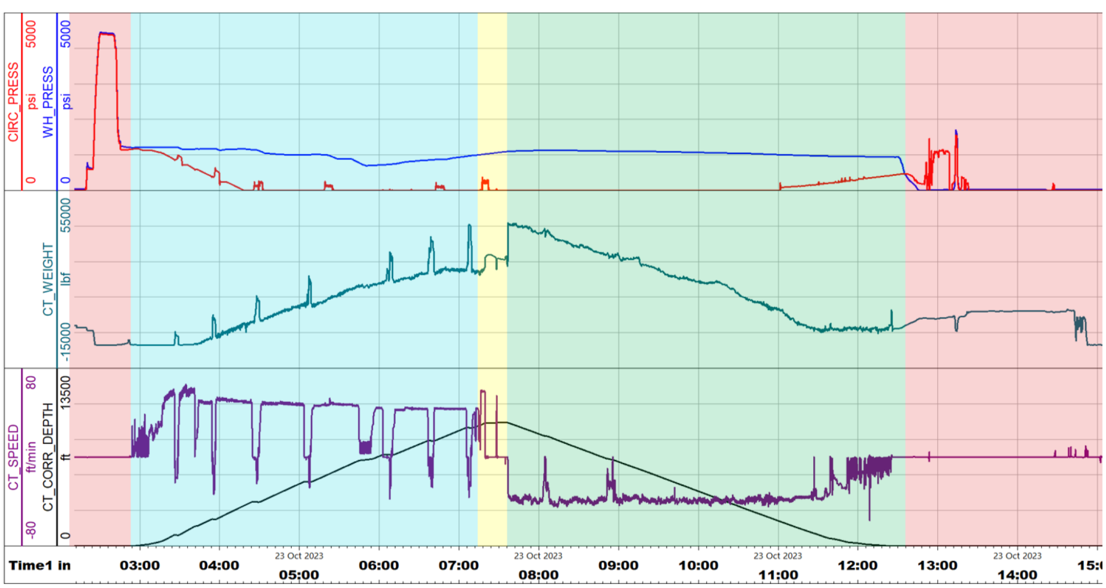

Figure 9—The above plot illustrates the cutting process. Red area highlights the pressure test of the equipment and the bled off and rig- down/rig-up of the cutterhead. Blue area depicts the RIH process. Yellow zone highlights the cutting process. Finally, the green zone represents the POOH process. One can notice the WHP is around 1000 psi, in average, during the entire process[1].

Case Study A - First Global Deployment of Rigless Needle Retrieval

The first field application of the rigless CT needle retrieval system was executed on a gas well completed with a 4-1/2" fishbone liner comprising **40 fishbone subs—**a total of **160 jetting needles**. The well targeted a **sour gas reservoir** containing **H2S and CO2**, requiring pressure control throughout the intervention, with wellhead pressure maintained at approximately **1,000 psi**.

Operational Sequence and Contingency Execution

The operation began with planned mechanical retrieval of the upper needles. Over **16 CT runs**, operators used a specially designed fishing BHA to **cut and capture needle tails** within the liner. Each run averaged around **10 hours**.

On **run #17**, CT encountered a hard tag, halting progress. The **predefined contingency plan** was activated:

1. **Run #18** - A **venturi junk basket** was deployed to remove any loose debris above the tagging point, avoiding misinterpretation in subsequent diagnostics.
2. **Run #19** - A **lead impression block (LIB)** was run to identify obstruction features. The LIB confirmed the presence of **nested or partially deployed needles**.
3. **Run #20** - As per plan, a **CT motor with flat mill** was mobilized for remedial action.

During **Run #21**, the mill tagged metal at **11,914 ft**, and was worked down to **12,175 ft**. A pull test and hang-off at **12,108 ft** prompted a decision to pull out of hole (POOH). However, during retrieval, the CT string **hung up at a side pocket mandrel** at **6,554 ft**.

After multiple unsuccessful attempts to pass the SPM, it was decided to continue **milling the horizontal section entirely** to regain access. Milling progressed down to **13,075 ft**, at 2.5 bpm using water and gel sweeps (Fig. 10).

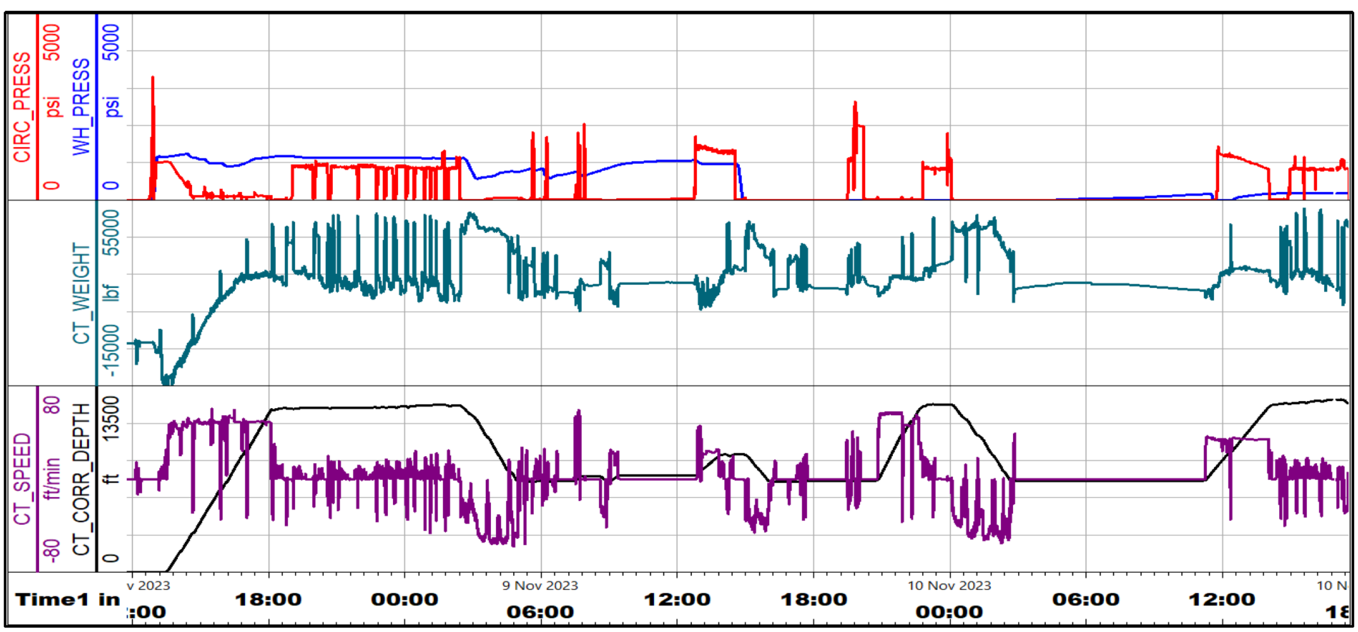

Figure 10—The above plot illustrates the milling run from case A[1].

Due to inability to retrieve the motor and mill, a **hydraulic disconnect** was activated, leaving the BHA safely in the rathole.

Drift Run and Final Recovery

A **blind box drift run** was conducted to push the fish to total depth. At **13,060 ft**, the CT encountered **~7,000 lb of slack-off** with no further advancement. A subsequent **increase in circulation pressure** confirmed the tool was at bottom. Overpull was observed around **10,360 ft**, prompting the team to pump **150 bbl of water** down the CT-tubing annulus to release the pipe.

Outcomes and Learnings

- **Total runs: 23**
- **Duration: 17 days**
- **Well pressure:** ~1,000 psi throughout (live-well conditions)
- **Upper needles recovered:** 99 out of 100 (from the top 25 subs)
- **Lower 60 needles:** Fully milled in-hole (from the bottom 15 subs)
- **Access to TD:** Successfully re-established
- **HSE incidents: Zero**

This operation represents the **first successful global implementation of a rigless, live-well needle retrieval system** using coiled tubing and a CT tower. Despite the complexity of the contingency execution, the intervention restored full access to the wellbore and provided valuable insight into needle penetration behavior and fishbone system performance in a sour gas environment.

Case Study B - Optimized Rigless Intervention with Enhanced Performance

The second deployment of the rigless CT needle retrieval system was carried out on a sour oil well completed with a 4-1/2" fishbone liner comprising **30 fishbone subs**, or **120 jetting needles**. The reservoir was characterized by **high CO2 content**, requiring strict pressure control and continuous circulation.

Operational Sequence and Adaptive Strategy

The operation began with a **drift run** to verify upper completion access. **Runs #2 to #16** were dedicated to cutting and retrieving the remaining needles using the CT tower and specialized fishing BHA.

On **run #16**, CT tagged hard at depth, indicating a potential obstruction. A fluid-only cleanout was initiated, but it became apparent the well could **not sustain pressure**, necessitating a shift to **nitrified cleanout operations** to avoid excessive hydrostatic loading.

A follow-up fishing run recovered **only two strainers and one needle**, confirming restricted access beyond **11,707 ft**. The contingency plan was activated:

1. A **venturi junk basket** was deployed to clean debris above the obstruction but returned empty.
2. A second-cutting attempt was made but again resulted in **tool hang-up at the same depth**.

After thorough risk assessment and considering well stability, the decision was made to **leave the remaining six fishbone subs (24 needles) in place** and proceed with upper completion installation.

Results and Operational Improvements

- **Needles recovered:** 92 out of 120 (from subs 30 to 7)
- **Remaining needles:** 28 from the lower 7 subs (milled or left in place)

- **Wellbore access:** Successfully re-established to 11,707 ft
- Runs completed: 16 • Intervention duration: ~14 days
- **Average efficiency:** Improved to ~2.5 **runs/day**, driven by enhanced **Pressure Control Equipment (PCE)** configuration and field execution, as illustrated in Fig 11.
- **HSE incidents: None**

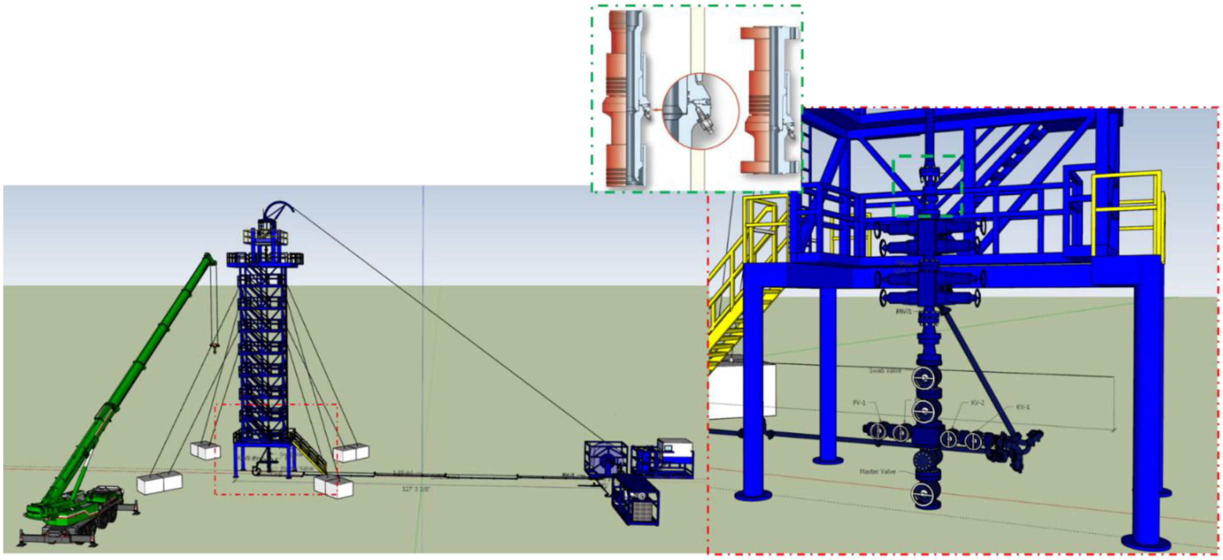

Figure 11—The improved CT Tower rigless approach, positioning the QTS at the lower end of the risers[1].

This case reflected the benefits of **iteration and optimization**. Learnings from the first well were applied to streamline tool handling, enhance pressure management, and accelerate cycle times. The improved intervention efficiency —achieved under live-well conditions—demonstrates the scalability of the **rigless CT tower methodology** for future applications.

Conclusions

This world-first implementation of a **rigless coiled tubing-based fishbone needle retrieval** method was successfully executed in two horizontal wells in the UAE, delivering substantial **operational, production, and economic benefits**.

In the **first well**, 40 fishbone subs were addressed, and in the **second**, 30 more, requiring an extended BHA to reach and recover the jetting needles. In total, **212 out of 240 needles were successfully retrieved**, enabling the operator to gain valuable insight into the effectiveness of the fishbone completion design (Alshehhi, 2025).

The intervention produced **exceptional results**:

- Case A: Gas production increased by 300%
- Case B: Oil production increased by 400%

These improvements exceeded operator expectations and were directly linked to the **removal of loss-control material**, which helped restore natural reservoir connectivity and reduce wellbore damage.

From an economic standpoint, the rigless method reduced the **completion-phase rig time from 44 days to just 10**, lowering **OPEX by an estimated 34%**. Combined with the production uplift, this innovation reduced the **return-on-investment period from 75 days to only 21.4 days**, accelerating value capture (Alshehhi, 2025).

Operationally, the rigless approach:

- Preserved **well integrity** by maintaining **well control throughout** the live intervention.
- Reduced **HSE exposure** by minimizing mechanical lifting and eliminating suspended loads—thanks to the CT tower consistently supporting the PCE.

Beyond operational success, this method established a **new benchmark** for post-completion interventions, offering the industry a safer, and more cost-effective **alternative** to conventional rig-based recovery operations. It also supports the UAE's broader objectives of **hydrocarbon production growth and resource optimization**.

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